

Standard Structure of Every Mathematical Induction Proof

- Base Case:** Verify the statement holds for the smallest value  $n = n_0$ .
- Inductive Hypothesis (IH):** Assume the statement holds for some arbitrary  $n = k \geq n_0$ .
- Inductive Step (IS):** Using the IH, prove the statement holds for  $n = k + 1$ .
- Conclusion:** By the Principle of Mathematical Induction (PMI), the statement holds for all  $n \geq n_0$ .

Question 1 | Prove:  $2^n > n^2$  for all  $n \geq 5$

TECHNIQUE — INEQUALITY INDUCTION

In the inductive step we double the IH and compare. We use the fact that for  $k \geq 5$ :  $2k^2 \geq (k + 1)^2$ , equivalent to  $k^2 - 2k - 1 \geq 0$ , which holds for  $k \geq 3$ .

STEP 1 — BASE CASE:  $N = 5$

$$\text{LHS} = 2^5 = 32 \quad \text{RHS} = 5^2 = 25$$

Since  $32 > 25$ , the base case holds. ✓

STEP 2 — INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 5$ :

$$2^k > k^2 \quad (\text{IH})$$

STEP 3 — INDUCTIVE STEP: SHOW  $2^{k+1} > (k+1)^2$

Starting from the LHS and applying the IH:

$$2^{k+1} = 2 \cdot 2^k \stackrel{\text{IH}}{>} 2k^2$$

Now show  $2k^2 \geq (k + 1)^2$  for  $k \geq 5$ :

$$2k^2 - (k + 1)^2 = k^2 - 2k - 1$$

For  $k \geq 5$ :  $k^2 - 2k - 1 \geq 25 - 10 - 1 = 14 > 0$ , so  $2k^2 > (k + 1)^2$ . ✓

Combining both inequalities:

$$2^{k+1} > 2k^2 \geq (k + 1)^2 \implies 2^{k+1} > (k + 1)^2$$

CONCLUSION

By the Principle of Mathematical Induction,  $2^n > n^2$  for all  $n \geq 5$ . ■

Question 2 | Prove:  $n^3 > 2n + 1$  for all  $n \geq 2$

STEP 1 – BASE CASE:  $N = 2$

$$\text{LHS} = 2^3 = 8 \quad \text{RHS} = 2(2) + 1 = 5$$

Since  $8 > 5$ , the base case holds. ✓

STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 2$ :

$$k^3 > 2k + 1 \quad (\text{IH})$$

STEP 3 – INDUCTIVE STEP: SHOW  $(K+1)^3 > 2(K+1) + 1$

Expand the LHS:

$$(k+1)^3 = k^3 + 3k^2 + 3k + 1$$

Apply the IH ( $k^3 > 2k + 1$ ):

$$(k+1)^3 > (2k+1) + 3k^2 + 3k + 1 = 3k^2 + 5k + 2$$

We need this to exceed  $2(k+1) + 1 = 2k + 3$ . Check the difference:

$$3k^2 + 5k + 2 - (2k + 3) = 3k^2 + 3k - 1$$

For  $k \geq 2$ :  $3k^2 + 3k - 1 \geq 12 + 6 - 1 = 17 > 0$ . ✓

$$\therefore (k+1)^3 > 2k + 3 = 2(k+1) + 1$$

CONCLUSION

By PMI,  $n^3 > 2n + 1$  for all  $n \geq 2$ . ■

Question 3 | Prove:  $3 \mid (n^3 - 7n + 3)$  for all  $n \geq 1$

DIVISIBILITY TECHNIQUE

Define  $P(n) = n^3 - 7n + 3$ . Expand  $P(k+1)$  and isolate  $P(k)$  so the IH can be applied directly. The remainder must be shown to also be divisible by 3.

STEP 1 – BASE CASE:  $N = 1$

$$P(1) = 1^3 - 7(1) + 3 = 1 - 7 + 3 = -3 = 3 \times (-1)$$

Since  $-3$  is divisible by 3, the base case holds. ✓

STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 1$ :

$$3 \mid (k^3 - 7k + 3) \iff k^3 - 7k + 3 = 3m \text{ for some integer } m \quad (\text{IH})$$

STEP 3 – INDUCTIVE STEP: SHOW  $3 \mid [(K+1)^3 - 7(K+1) + 3]$

Expand  $P(k+1)$ :

$$\begin{aligned}(k+1)^3 - 7(k+1) + 3 &= k^3 + 3k^2 + 3k + 1 - 7k - 7 + 3 \\ &= \underbrace{(k^3 - 7k + 3)}_{\text{IH: divisible by 3}} + 3k^2 + 3k - 6 \\ &= (k^3 - 7k + 3) + 3(k^2 + k - 2)\end{aligned}$$

$3 \mid (k^3 - 7k + 3)$  by the IH, and  $3 \mid 3(k^2 + k - 2)$  trivially. The sum of two multiples of 3 is divisible by 3.

$$\therefore 3 \mid [(k+1)^3 - 7(k+1) + 3] \checkmark$$

CONCLUSION

By PMI,  $3 \mid (n^3 - 7n + 3)$  for all  $n \geq 1$ . ■

Question 4 | Prove:  $1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$  for all  $n \geq 1$

GEOMETRIC SERIES

The sum  $\sum_{i=0}^{n-1} 2^i$  has  $n$  terms:  $2^0, 2^1, \dots, 2^{n-1}$ . The next term when moving from  $n = k$  to  $n = k + 1$  is  $2^k$ .

STEP 1 – BASE CASE: N = 1

$$\text{LHS} = 2^0 = 1 \qquad \text{RHS} = 2^1 - 1 = 1$$

LHS = RHS. ✓

STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 1$ :

$$1 + 2 + 4 + \dots + 2^{k-1} = 2^k - 1 \qquad \text{(IH)}$$

STEP 3 – INDUCTIVE STEP: SHOW THE SUM TO  $2^k$  EQUALS  $2^{k+1} - 1$

Add the next term  $2^k$  to both sides of the IH:

$$\begin{aligned}1 + 2 + \dots + 2^{k-1} + 2^k &= (2^k - 1) + 2^k \\ &= 2 \cdot 2^k - 1 \\ &= 2^{k+1} - 1\end{aligned}$$

This is exactly the statement for  $n = k + 1$ . ✓

CONCLUSION

By PMI,  $1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$  for all  $n \geq 1$ . ■

Question 5 | Prove:  $11 \mid (23^n - 1)$  for all  $n \geq 0$

KEY OBSERVATION – WHY THIS WORKS

$23 = 22 + 1 = 2(11) + 1$ , so  $23 \equiv 1 \pmod{11}$ . Therefore  $23^n \equiv 1^n = 1 \pmod{11}$  for all  $n \geq 0$ , giving  $23^n - 1 \equiv 0 \pmod{11}$ . The induction proof below formalises this using the substitution  $23^k = 11m + 1$ .

STEP 1 – BASE CASE: N = 0

$23^0 - 1 = 1 - 1 = 0 = 11 \times 0$

Since 0 is divisible by 11, the base case holds. ✓

STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 0$ :

$11 \mid (23^k - 1) \iff 23^k - 1 = 11m \text{ for some integer } m \tag{IH}$

Equivalently:  $23^k = 11m + 1$ .

STEP 3 – INDUCTIVE STEP: SHOW  $11 \mid (23^{k+1} - 1)$

Write  $23^{k+1} - 1$  and substitute the IH:

$$\begin{aligned} 23^{k+1} - 1 &= 23 \cdot 23^k - 1 \\ &= 23(11m + 1) - 1 \quad \text{[using } 23^k = 11m + 1 \text{ from IH]} \\ &= 253m + 23 - 1 \\ &= 253m + 22 \\ &= 11(23m + 2) \end{aligned}$$

Since  $(23m + 2)$  is an integer,  $23^{k+1} - 1 = 11(23m + 2)$  is divisible by 11. ✓

**Verification with small values:**  $n = 1 : 23^1 - 1 = 22 = 11 \times 2 \checkmark \mid n = 2 : 23^2 - 1 = 528 = 11 \times 48 \checkmark \mid$   
 $n = 3 : 23^3 - 1 = 12166 = 11 \times 1106 \checkmark$

CONCLUSION

By the Principle of Mathematical Induction,  $11 \mid (23^n - 1)$  for all  $n \geq 0$ . ■

Question 6 | Prove:  $1 + 3 + 5 + \dots + (2n - 1) = n^2$  for all  $n \geq 1$

SUM OF THE FIRST N ODD NUMBERS

The  $k$ -th odd number is  $2k - 1$ . The next odd number (the  $(k + 1)$ -th) is  $2(k + 1) - 1 = 2k + 1$ .

STEP 1 – BASE CASE: N = 1

$LHS = 2(1) - 1 = 1 \quad RHS = 1^2 = 1$

LHS = RHS. ✓

STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 1$ :

$1 + 3 + 5 + \dots + (2k - 1) = k^2 \tag{IH}$

STEP 3 – INDUCTIVE STEP: SHOW THE SUM TO  $(2K+1)$  EQUALS  $(K+1)^2$

Add the next odd number  $(2k + 1)$  to both sides of the IH:

$$\begin{aligned} 1 + 3 + \dots + (2k - 1) + (2k + 1) &= k^2 + (2k + 1) \\ &= (k + 1)^2 \end{aligned}$$

This is exactly the statement for  $n = k + 1$ . ✓

CONCLUSION

By PMI,  $1 + 3 + 5 + \dots + (2n - 1) = n^2$  for all  $n \geq 1$ . ■

Question 7 | Prove:  $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n + 1)^2}{4}$  for all  $n \geq 1$

KEY ALGEBRAIC MOVE IN THE INDUCTIVE STEP

After adding  $(k + 1)^3$  to the IH, factor  $\frac{(k + 1)^2}{4}$  from the resulting expression. The bracket simplifies as  $k^2 + 4(k + 1) = k^2 + 4k + 4 = (k + 2)^2$ .

STEP 1 – BASE CASE: N = 1

$$\text{LHS} = 1^3 = 1 \quad \text{RHS} = \frac{1^2 \cdot 2^2}{4} = \frac{4}{4} = 1$$

LHS = RHS. ✓

STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 1$ :

$$1^3 + 2^3 + \dots + k^3 = \frac{k^2(k + 1)^2}{4} \quad (\text{IH})$$

STEP 3 – INDUCTIVE STEP: SHOW THE SUM TO  $(K + 1)^3$  EQUALS  $\frac{(K + 1)^2(K + 2)^2}{4}$

Add  $(k + 1)^3$  to both sides of the IH:

$$1^3 + \dots + k^3 + (k + 1)^3 = \frac{k^2(k + 1)^2}{4} + (k + 1)^3$$

Factor  $\frac{(k + 1)^2}{4}$  from the right side:

$$\begin{aligned} &= \frac{(k + 1)^2}{4} [k^2 + 4(k + 1)] \\ &= \frac{(k + 1)^2}{4} [k^2 + 4k + 4] \\ &= \frac{(k + 1)^2(k + 2)^2}{4} \end{aligned}$$

This is the formula evaluated at  $n = k + 1$ . ✓

CONCLUSION

By PMI,  $\sum_{i=1}^n i^3 = \frac{n^2(n + 1)^2}{4}$  for all  $n \geq 1$ . ■

Question 8 | Prove:  $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1}n^2 = \frac{(-1)^{n-1}n(n + 1)}{2}$  for all  $n \geq 1$

ALTERNATING SIGN – KEY IDENTITY USED

$(-1)^k = -(-1)^{k-1}$ . When moving from step  $k$  to  $k + 1$ , the sign on the RHS flips. This identity is essential for simplifying the expression after adding the next term.

### STEP 1 – BASE CASE: N = 1

$$\text{LHS} = (-1)^0 \cdot 1^2 = 1 \quad \text{RHS} = \frac{(-1)^0 \cdot 1 \cdot 2}{2} = 1$$

LHS = RHS. ✓

**Sanity check at n = 2:**

$$\text{LHS} = 1 - 4 = -3 \quad \text{RHS} = \frac{(-1)^1 \cdot 2 \cdot 3}{2} = -3 \checkmark$$

### STEP 2 – INDUCTIVE HYPOTHESIS

Assume for some  $k \geq 1$ :

$$1^2 - 2^2 + \dots + (-1)^{k-1} k^2 = \frac{(-1)^{k-1} k(k+1)}{2} \quad (\text{IH})$$

### STEP 3 – INDUCTIVE STEP: SHOW THE SUM TO $(-1)^K(K+1)^2$ EQUALS $\frac{(-1)^K(K+1)(K+2)}{2}$

Add the next term  $(-1)^k(k+1)^2$  to both sides of the IH:

$$\text{LHS}_{k+1} = \frac{(-1)^{k-1} k(k+1)}{2} + (-1)^k(k+1)^2$$

Factor out  $(-1)^{k-1}(k+1)$ , using  $(-1)^k = -(-1)^{k-1}$ :

$$\begin{aligned} &= (-1)^{k-1}(k+1) \left[ \frac{k}{2} - (k+1) \right] \\ &= (-1)^{k-1}(k+1) \left[ \frac{k - 2(k+1)}{2} \right] \\ &= (-1)^{k-1}(k+1) \left[ \frac{-k - 2}{2} \right] \\ &= (-1)^{k-1}(k+1) \cdot \frac{-(k+2)}{2} \end{aligned}$$

Since  $(-1)^{k-1} \times (-1) = (-1)^k$ :

$$= \frac{(-1)^k(k+1)(k+2)}{2}$$

This is the formula evaluated at  $n = k + 1$ . ✓

### CONCLUSION

By PMI,  $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = \frac{(-1)^{n-1} n(n+1)}{2}$  for all  $n \geq 1$ . ■